Analysis Choices and Ranking Stability in Kinase Inhibitor Selectivity Scores

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Abstract

Across four kinase profiling datasets, affinity weighting, floor-coded pairs and compound eligibility alter agreement between selectivity scores. On random smaller panels, rank agreement with the full panel can remain high even as compounds with few active targets become unscorable. Moving the activity cutoff changes rankings, so it should be fixed by protocol. We introduce four design criteria. Within a specified class of affinity-weighted scores, these criteria admit only gated entropies with exponential or equal weights. We derive the scale range that prevents added weak binding from increasing apparent selectivity. On the datasets studied, the resulting score has no recovery advantage over simpler scores. We recommend that comparisons should report how scores are calculated, which compounds can be scored, sensitivity to the activity cutoff, and how many compounds remain scorable.

Keywords: binding selectivity, chemoinformatics, drug discovery, kinase inhibition, ranking stability.

Introduction

Kinase inhibitors are among the most investigated compound classes in drug discovery, with 94 agents approved by the U.S. Food and Drug Administration (FDA) and several hundred in clinical development[1]. The human kinome is comprised of more than 500 proteins with a conserved adenosine triphosphate (ATP)-binding site[2], and selectivity is a primary criterion in lead optimization[3]. Profiling assays measure a compound against a panel of tens to hundreds of kinases, and the resulting profile is often reduced to one selectivity score. Several scores are in use:

- the S-score[4], the fraction of the panel bound below an affinity cutoff
- the selectivity entropy[5], the Shannon entropy of the profile treated as a distribution
- the Gini coefficient[6] of the profile
- the partition index[7], the share of binding that falls on one target
- ratio-based measures, which compare the primary target with an off-target.

They are often used interchangeably, although each summarizes a different aspect of the profile.

Earlier work established that these scores can disagree. Bosc et al.[8] compared six scores on the Davis and Anastassiadis datasets, computed several at more than one parameter value, then compared them on compounds and kinases shared between datasets. They found that compound rankings differ substantially between them. They attributed individual disagreements to profile shape, such as a broad binder with one dominant target. The KInhibition platform[9] noted that conflicting results across metrics are common, and introduced a composite score. A later resource[10] from the same group offers a choice of scores. Miljkovic and Bajorath[11] found the concentration-dependent selectivity score of Klaeger et al.[12] broadly consistent with activity data from ChEMBL. Karaman et al.[4] and Uitdehaag et

al.[13] showed that the S-score and the entropy of individual compounds vary when they are computed on random subpanels.

These studies establish disagreement between scores and sensitivity to parameters and panel composition. We examine how score agreement changes when affinity weighting, floor-coded pairs, compound eligibility and target selection are varied explicitly. On smaller panels, we assess both how well compounds that remain scorable retain their full-panel ranking and what fraction of the full panel's scorable compounds can still be scored. Rank agreement can remain high even when compounds with few active targets become unscorable.

Our primary contribution is an empirical comparison of these analysis choices, showing how ranking recovery on smaller panels depends on which compounds remain scorable. We compare the S-score, entropy, Gini and ratio on four datasets that span three assay technologies. Each is curated to one reference target per human protein kinase, and the entropy and Gini are compared in baseline-subtracted and published forms where the required readout is available. We vary the compounds ranked, the treatment of floor-coded pairs, and each score's parameter, then relate the resulting instability to features of the binding profile as exploratory diagnostics. Motivated by these sensitivities, we propose and defend four design and reporting criteria for a selectivity measure: activity eligibility, locality of the activity boundary, monotonicity when a weakly bound kinase is added, and orientation on profiles with equally strong targets. We evaluate the existing scores and simple comparators against them.

Our second contribution is a conditional characterization of gated entropies that satisfy all four. We consider scores that weight each target according to its distance above the activity boundary. We require the same result whether targets are evaluated all at once or in groups: the score across groups plus the scores within each group, weighted by each group's share of the total weight. Under these assumptions, the criteria allow only exponential weights (including the limiting case in which every target above the boundary receives the same weight). At finite scales, the exponential family has a single parameter, the *scale*, the readout

difference over which a weight grows by a factor of e. Monotonicity fixes the range of the scale. Within that range, the scale can be chosen so that the score recovers rankings from small panels as fast as the share of panel entries above the operational floor, but an empirical recovery advantage over existing measures is not identified.

Methods

Datasets, Target Selection and Missing Values

Table 1 summarizes the four datasets. The Davis dataset [14] reports dissociation constants from a competition-binding assay. The Klaeger dataset [12] reports apparent dissociation constants from Kinobeads chemoproteomics in cell lysate. Kinobeads carry kinase inhibitors that capture kinases from the lysate, and mass spectrometry measures the captured proteins. With constants in nM, we converted both datasets, as in DeepDTA [15], to $\mathrm{p}K_d = -\log_{10}(K_d/10^9)$. The Anastassiadis dataset [16] reports residual kinase activity in a radiometric assay at 0.5 μM inhibitor. We converted this to percent inhibition (100 minus residual activity, clipped to $[0, 100]$). The Metz dataset [17] reports $\mathrm{p}K_i$ values from a medicinal-chemistry campaign. Because the assays and readouts differ, we compute each score on its dataset's own readout. We compare datasets through the resulting rankings.

Table 1: The four datasets. The floor x_0 is the value given to pairs with no detected binding (Davis, Klaeger) or not tested (Metz). On Anastassiadis it is a chosen cutoff for weak inhibition. τ^* is the activity threshold.

Dataset	Assay; readout	Compounds	Targets (released, curated)	x_0	τ^*
Davis	competition binding; $\mathrm{p}K_d$	68	433, 363	5.0	6
Klaeger	Kinobeads in lysate; $\mathrm{p}K_d$	222	343, 217	5.0	6
Anastassiadis	radiometric activity; % inhibition	178	300, 291	20%	50%
Metz	biochemical activity; $\mathrm{p}K_i$	704	172, 172	4.0	6

Target selection. A kinase may appear several times in a profile as there may be multiple constructs of one gene. Constructs are recombinant forms of a protein, such as mutants

or single domains. The released Davis panel lists 433 constructs. These include mutant, phosphorylation-state and single-domain forms of the same gene (12 for EGFR), 20 lipid kinases and three non-human kinases. For 26 of the 68 compounds, the two best-bound constructs are forms of one gene. We keep one reference construct per human protein-kinase gene: the wild-type construct, or its catalytic or N-terminal kinase domain where several domains are profiled. Genes profiled only as mutants are removed.

The 343 Klaeger targets are proteins captured by the beads and measured by mass spectrometry. They include complex subunits, such as cyclins, that bind the captured kinases. We keep the targets that Klaeger et al. annotate as single-gene protein kinases, except Q6ZSR9. UniProt lists Q6ZSR9 as an uncharacterized protein without a kinase domain. This leaves 217 targets and removes lipid kinases, four protein groups, BCR and TRRAP. The source does not annotate BCR and TRRAP as kinases and flags them as possibly co-enriched with ABL1 and CDK8.

On Anastassiadis, nine kinases are profiled in two complexes or isoforms. We keep the first-listed form of each. The Metz columns already represent one human protein kinase each. We call the targets kept in each dataset its *curated panel* (Table 1). The code repository lists each target and the reason it was kept or removed. Supporting Information (Section S1) compares results on the released panels.

Floor-coded and missing values. On Davis, Klaeger and Metz, several kinds of pairs receive the same floor value x_0. This value is not a measured detection limit. On Davis, the released matrix assigns $\mathrm{p}K_d = 5.0$ ($K_d = 10$ μM) to pairs with no detected binding, following DeepDTA[15]. Bosc et al.[8] used nearly the same value (10.1 μM).

On Klaeger, we kept the 5,123 drug–target pairs that the original authors classified as high confidence. We assigned $\mathrm{p}K_d = 5.0$ to three kinds of pairs: those with no detected binding, those whose fit was classified as low confidence, and the 194 measured affinities weaker than 10 μM. A low-confidence fit does not establish weak binding: of the 405 such

pairs on curated targets, 375 have a fitted pK_d above 5. Giving all of these pairs the floor value is an analysis convention.

On Metz, pairs that were not tested are coded at p$K_i = 4.0$, the lowest value reported in the data. The median number of kinases measured per compound is only 21 out of 172. We therefore kept the 704 compounds measured against at least 50 kinases (median 64).

On Anastassiadis, the 1.1% of missing entries were set to 0% inhibition. Most pairs on this panel show little inhibition: 85% of the entries are at or below 20%, and each 5% interval above 20% holds less than 3% of the entries. We place x_0 at 20%, the upper end of these low values, as a chosen cutoff for weak inhibition.

A compound is *active* against a target if its readout exceeds the activity threshold τ^*. On the affinity datasets, this threshold is 6 on the pK_d or pK_i scale (K_d or K_i of 1 μM). On Anastassiadis, it is 50% inhibition. With a Hill coefficient of 1, 50% inhibition at 0.5 μM inhibitor means an IC_{50} of 0.5 μM, close to the 1 μM of the other datasets. Supporting Information (Section S1) repeats the Davis comparison of the scores with the threshold one log unit higher. An *active compound* has at least one target above τ^*.

Selectivity Scores and Their Implementations

A compound's profile is the vector $\mathbf{x} = (x_1, \ldots, x_N)$ of its readouts across the N panel targets, with higher values for tighter binding or stronger inhibition. Every score is oriented so that higher values are more selective (Table 2).

S-score. Karaman et al.[4] define the S-score as the fraction of the panel bound above a threshold:

$$S(\tau) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[x_i > \tau]. \tag{1}$$

We negate this fraction so that higher scores indicate greater selectivity. They reported K_d cutoffs of 3 μM and 100 nM, which lie within our sweep of τ.

Selectivity entropy. The selectivity entropy of Uitdehaag and Zaman[5] is the Shannon entropy $H = -\sum_i p_i \log_2 p_i$ (in bits) of the profile normalized to a distribution. We negate it so that higher scores indicate greater selectivity. In its published form, $p_i = K_{a,i} / \sum_j K_{a,j}$, with association constants $K_{a,i} = 1/K_{d,i}$ ($1/K_{i,i}$ on Metz). It needs affinity constants, so we do not compute it on Anastassiadis. To compare all four definitions on the same readout, we also use a log-affinity form. This weights each target by its readout above a baseline β. On Anastassiadis, the readout is percent inhibition:

$$p_i = \frac{\max(x_i - \beta,\, 0)}{\sum_j \max(x_j - \beta,\, 0)}. \tag{2}$$

We introduce this baseline parameter, which has no published reference values.

Gini coefficient. Graczyk's Gini coefficient[6] measures how unequally binding is spread over the panel. Its published form uses percent-inhibition values. We compute it on the same baseline-subtracted weights:

$$G(\beta) = \frac{2\sum_{i=1}^{N} i \cdot w_{(i)}}{N \sum_{i=1}^{N} w_{(i)}} - \frac{N+1}{N}, \qquad w_{(i)} = \max(x_i - \beta,\, 0) \text{ sorted ascending.} \tag{3}$$

On Anastassiadis, we also compute the published percent-inhibition form. On Davis and Klaeger, we also compute the association-constant Gini (K_a-Gini) introduced by Uitdehaag and Zaman[5]. This variant applies the Gini formula to K_a values rather than percent inhibition. Figure 1 compares it with the baseline-subtracted form.

Unless stated otherwise, both the entropy and Gini are in their log-affinity forms.

Ratio. The ratio compares the primary target with the k-th ranked off-target,

$$R_b(k) = x_{(1)} - \max(x_{(k+1)}, b), \tag{4}$$

where $x_{(j)}$ is the j-th largest value. The level b is the lowest coded value (5.0 on Davis and Klaeger, 4.0 on Metz and 0% on Anastassiadis). In the recovery analysis, we instead set it to x_0. In the boundary analysis, we vary it over the sweep. At the lowest coded value, $R_b(k)$ equals the ranking score of Bosc et al.[8] at rank $k+1$. We cap k at 5 because half of the Klaeger compounds have fewer than five active kinases.

Partition index. The partition index of Cheng et al.[7] is the fraction of a compound's molecules that would bind a reference target in a mixture of all panel kinases, $K_{a,\mathrm{ref}} / \sum_j K_{a,j}$. Only Klaeger names an intended target for each compound. We therefore use the most strongly bound target as the reference ($P_{\max}$), as Uitdehaag and Zaman[5] did. Like the published entropy, it is not computed on Anastassiadis.

Klaeger et al.'s concentration- and target-dependent selectivity (CATDS) score is the same kind of fraction, for a designated target at one concentration. It requires Kinobeads dose–response curves, which the other three datasets lack. We therefore do not compare it across datasets. Supporting Information (Section S6) compares a proxy.

Table 2: Score implementations. Parameter ranges are for the pK_d datasets. On Anastassiadis τ runs over $[50, 90]$% and β over $[0, 30]$%, and on Metz over $[5.0, 7.5]$ and $[4.0, 5.5]$. The last three rows are comparators.

Score	Weights or formula	Parameter swept	No target above the parameter
S-score	fraction above τ	$\tau \in [5.5, 8.0]$	score 0
Entropy, log-affinity	$\max(x_i - \beta, 0)$	$\beta \in [5.0, 6.5]$	undefined; set to 0 (top rank)
Gini, log-affinity	$\max(x_i - \beta, 0)$	$\beta \in [5.0, 6.5]$	undefined; set to 0 (bottom rank)
Ratio $R_b(k)$	$x_{(1)} - \max(x_{(k+1)}, b)$	$k = 1, \ldots, 5$	uses b
Entropy, published	$K_{a,i}$	none	—
Partition index $P_{\max}$	$\max_i K_{a,i} / \sum_j K_{a,j}$	none	—
Gated entropy	gated exponential (below)	none	not scored (D1)
Active count	number above τ^*	none	not scored (D1)
Entropy, $\beta = \tau^*$	$\max(x_i - \tau^*, 0)$	none	not scored (D1)
S-score, $\tau = x_0$	fraction above x_0	none	not scored (D1)

Conventions. Where the entropy and Gini are undefined, we set both to 0. This ranks the compound as maximally selective under the entropy and minimally selective under the Gini. $R_b(k)$ uses b as a lower bound on the reference off-target value and scores every compound, as Bosc et al.'s ranking score does. Log-affinity weights with $\beta = x_0$ give floor-coded pairs zero weight. Association constants give them a small positive weight if they are kept at the floor value. For the published entropy and $P_{\max}$, we give them zero weight. As a sensitivity analysis for the published entropy, we keep floor-coded pairs at the floor value. On Klaeger, we also give every measured high-confidence affinity its measured value, including the 194 weaker than the floor, and the other floor-coded pairs zero weight. No data show whether floor-coded pairs should have zero weight or the floor value.

Comparators. We use three variants of existing scores. The active count is the number of targets above τ^*, or the numerator of the S-score at τ^*. A lower count indicates greater selectivity. The other two comparators are the entropy with $\beta = \tau^*$ and the S-score with $\tau = x_0$, the share of panel entries above the operational floor. On Davis and Klaeger, these are retained affinities stronger than the coded floor; on Anastassiadis, inhibition above 20%; and on Metz, almost every measured entry. This share therefore has a different interpretation on each dataset. We compute all three only for active compounds. Otherwise, a compound with no active target would have a count of 0 and rank as more selective than any active compound.

Ranking Comparisons and Sensitivity Analysis

For each score and parameter value, compounds are ranked from most selective (rank 1), with ties given their average rank. Analyses of compounds with an active kinase rank them among themselves. Analyses that compare compounds with and without an active kinase rank all compounds. Agreement between two definitions is the Spearman correlation between their median rank vectors. We take the median over each definition's parameter sweep. On

the pK datasets, we vary τ in steps of 0.25 (11 values) and β in steps of 0.25 (7 values). On Anastassiadis, we vary τ in steps of 5% (9 values) and β in steps of 5% (7 values). For the ratio, we use $k = 1, \ldots, 5$. Bootstrap 95% intervals use 1,000 resamples of compounds.

A compound's *rank instability* is the standard deviation of its ranks across one definition's sweep. Its *aggregated rank instability* is the standard deviation of its ranks across all four sweeps combined. Each definition therefore counts in proportion to its number of parameter values, and the S-score, with the most values, carries the most weight. We compute the Spearman correlation between within-definition instability and each of six profile features. These are the number of active kinases, the top_1–top_2 affinity gap, the second-ranked affinity, and the standard deviation, range and skewness of the active affinities. A feature undefined for a compound with too few active kinases is set to 0. Supporting Information (Section S3) repeats the analysis with these compounds excluded. These associations are exploratory. We apply a Bonferroni correction to the 24 tests per dataset in Supporting Information Table S4: six features for each of four definitions. We also check the Holm and Benjamini–Hochberg procedures.

We also compare changes in ranking between datasets with changes between definitions. We matched 46 compounds profiled in both Davis and Klaeger by PubChem compound identifier (CID). We restricted both panels to their 201 shared kinases, matched by gene symbol after updating renamed symbols. We then computed two sets of Spearman correlations: between each definition's median rankings on Davis and Klaeger, and between pairs of definitions' median rankings within each dataset.

For these 46 compounds, we also tested whether cross-dataset rank displacement was associated with instability aggregated over all four definitions or calculated within the same definition's parameter sweep. These exploratory analyses used each dataset's full curated target panel and original compound ranking population. We apply a separate Bonferroni correction to the eight within-definition tests: four definitions in each of two datasets. Other p-values are uncorrected.

Evaluation Criteria

A selectivity measure f assigns a real number $f(\mathbf{x})$ to a profile $\mathbf{x}$ of any length N. Higher values indicate greater selectivity. Let $B = [x_0, \tau^*]$ be the interval between the floor and the activity threshold. D1 decides which compounds are scored. D2 concerns moving the activity boundary, and D3 concerns adding targets to the panel. D4 concerns the direction of the score. We judge all four from the score definitions.

D1 (activity eligibility). $f(\mathbf{x})$ is undefined, or flagged as ineligible, whenever $\max_i x_i \leq \tau^*$. A compound with no target above τ^* has no target against which to judge selectivity. D1 is a reporting rule: a score satisfies it if its definition withholds or flags these compounds. Bosc et al.[8] withheld ranks for such compounds under the S-score only, whereas D1 applies the rule to every score.

D2 (boundary locality). Let θ be the score's boundary parameter: the S-score's τ, the entropy's and Gini's β, the ratio's level b, or the gated entropy's gate g. With all other parameters held fixed, D2 requires

$$\{i : x_i > \theta\} = \{i : x_i > \theta'\} \quad \Longrightarrow \quad f(\mathbf{x}; \theta) = f(\mathbf{x}; \theta').$$

Moving the boundary may change a compound's score only if it changes which targets lie above it. D2 concerns the score itself, not just the compound's rank. We keep D1's eligibility threshold τ^* fixed as θ changes.

D2 allows rankings to change when the boundary crosses a target's measured value. To measure these changes, we move θ across B. A pair of compounds *reverses in* B if $f(\mathbf{x}; \theta) - f(\mathbf{y}; \theta)$ is positive at one value of $\theta \in B$ and negative at another. We report the fraction of pairs that reverse among compounds eligible under D1.

For the S-score, we check every observed value in B, values between consecutive observations and the endpoints. These checks capture every score change, so the reversal percentages are exact. For the entropy, Gini and ratio, we use 201 equally spaced boundary values. Their

reversal percentages are estimates from this grid. For the gated entropy, we also check values at and just below each jump, as described below.

Some scores use a fixed boundary. For example, the active count counts targets above τ^*. We assess its sensitivity by varying the counting threshold over B, while keeping D1's eligibility rule fixed.

The analysis protocol should set θ in advance to suit the intended use. For example, binding weaker than τ^* may matter at the concentration used in practice. Different boundary values can address different questions. On Davis, the S-score at $\tau = 5$ counts targets with $K_d < 10$ μM. At $\tau = 6$, it counts those with $K_d < 1$ μM. Its rankings can reasonably differ between these cutoffs.

D3 (monotonicity under weak-target addition). For any $\mathbf{x}$ with $\max_i x_i > \tau^*$, if $\mathbf{x}' = (x_1, \ldots, x_N, x_{N+1})$ with $x_0 < x_{N+1} \leq \tau^*$, then $f(\mathbf{x}') \leq f(\mathbf{x})$. Otherwise a compound can be made to look more selective by profiling it against more kinases it binds weakly. Tests on measured profiles (Results) provide secondary checks.

D4 (orientation). Fix the panel size $N \geq 2$ and all of the score's parameters. For every value a above τ^* and above every cutoff and baseline the score uses, let $\mathbf{v}_m$ be the profile with m targets at a and the other $N - m$ at the floor, for $m = 1, \ldots, N$. D4 requires $f(\mathbf{v}_{m+1}) \leq f(\mathbf{v}_m)$ for $m = 1, \ldots, N-1$, and $f(\mathbf{v}_1) > f(\mathbf{v}_N)$. These conditions set the direction of the score. On the same panel, binding equally strongly to more targets must not make a compound look more selective. Binding to one target must also be more selective than binding equally strongly to all. The score need not decrease at every step: $R_b(k)$ with $1 \leq k < N$ equals $a - \max(x_0, b)$ for $m \leq k$ and 0 for $m > k$.

When floor-coded pairs have zero weight, the profiles $\mathbf{v}_m$ become both broader and more evenly weighted as m increases. When they keep a positive weight, they form a background of small weights. Turning them into equally strong targets moves weight from this background into a few large weights. The distribution can then become less even and the score can rise. A score can therefore fail D4 because of how it treats floor-coded pairs. We assess

each implementation with its own treatment (Supporting Information, Section S4, gives an example).

D4 constrains only the profiles $\mathbf{v}_m$. On a profile with unequal affinities, adding an active target that binds much more strongly than the others moves most of the weight to it. A score that measures how concentrated binding is, such as the entropy or $P_{\max}$, then rates the compound as more selective (Supporting Information, Section S4). Whether this rating is correct depends on what selectivity means: breadth (how many targets are bound) or concentration (how much binding falls on a few targets). D4 leaves this question open.

We state no criterion for the ratio's choice of reference off-target, because different references answer different questions.

The Gated Entropy

For a profile with $\max_i x_i > \tau^*$ (undefined otherwise, as D1 requires), define

$$w_i = \mathbf{1}[x_i > g]\, e^{(x_i - g)/T}, \qquad p_i = \frac{w_i}{\sum_j w_j}, \qquad f(\mathbf{x}) = \sum_i p_i \log_2 p_i, \tag{5}$$

with the gate $g = x_0$ and the scale $T = \tau^* - x_0$. The scale is the readout difference over which a weight grows by a factor of e. It is 1 pK unit on Davis and Klaeger, 2 on Metz, and 30 percentage points on Anastassiadis. The score is the negative Shannon entropy of a gated, exponentially weighted profile. The gate gives floor-coded pairs zero weight, and a kinase just above the gate has weight close to 1. Under the entropy's hard cutoff $\max(x_i - \beta, 0)$, such a kinase has weight close to 0. On pK data the weights are powers of the association constants, $w_i \propto K_{a,i}^{1/(T \ln 10)}$. The published entropy with floor-coded pairs given zero weight has the same form with $T = 1/\ln 10 \approx 0.43$ pK units. Moving the gate without passing a value multiplies every weight above it by the same factor. This factor cancels in the shares, so the gated entropy satisfies D2. A reference value subtracted from every readout cancels in the same way, so the score has no baseline parameter. The score jumps when the gate passes

an observed value. To count reversals, we therefore check each observed value in B, a value just below each one, and the 201 equally spaced values.

Proposition 1 (monotonicity). Adding a kinase with a value in $(x_0, \tau^*]$ never raises f if and only if $T \geq \tau^* - g$. With $T = \tau^* - x_0$, this holds for every gate $g \in B$.

The proof (Supporting Information, Section S4) has two parts. For the *if* part, every gated target has weight above 1, and the added target has weight 0 (at or below the gate) or at most $e^{(\tau^*-g)/T} \leq e$. If every weight is at least 1, appending one weight no larger than e cannot lower the entropy. For the *only if* part, for any smaller T, a profile with one active target and many kinases just above the gate scores higher after a kinase is added at τ^*.

Proposition 1 leaves T free above $\tau^* - x_0$. Two targets whose readouts differ by Δx have weights in the ratio $e^{\Delta x/T}$, so a smaller T makes the score depend more on affinity differences. We take the smallest T that D3 allows, $T = \tau^* - x_0$. Results ("Scale of the Gated Entropy and Recovery") compares $T = 2(\tau^* - x_0)$.

Because a target at the gate has weight 0 and any target above it has weight above 1, the score jumps when a target's value crosses x_0. For example, on Davis or Klaeger ($x_0 = 5$, $T = 1$), take a profile with one active target at $\mathrm{p}K_d = 7$. Adding a second target at the floor leaves the score at 0. Adding one just above the floor lowers it by 0.53 bits. Measurement noise near the floor could therefore change scores.

To measure this effect, we added normal noise to measured values above the floor, using standard deviations of 0.1 and 0.3 pK units. We set values pushed below the floor to the floor and left floor-coded pairs fixed. On Anastassiadis, we added noise to every entry, using standard deviations of 3 and 9 percentage points. Each setting used 50 draws. Noise can also change which compounds are scored: a compound whose strongest target lies just above τ^* can fall below it, and the reverse. D1 was therefore applied to each noisy profile. We compared rankings among compounds scored both with and without noise.

Proposition 2 (characterization). Let a score give each target above the gate g the weight $h(x_i - g)$ and give the other targets zero weight. Here h is positive, continuous,

nondecreasing and the same for every target. Let the score have the form $-F(\mathbf{p})$ of the weight shares $\mathbf{p}$. Require F to be continuous and to have Shannon's grouping property [18]. This means that, when kinases are split into groups, F equals its value across the groups plus its values within them, weighted by each group's share. If the score satisfies D2 and D4, then h is constant or $h(t) = Ce^{t/T}$ for some $C > 0$ and $T > 0$. The score is a positive multiple of the gated entropy with that T, or with equal weights if h is constant.

The proof (Supporting Information, Section S4) has two steps. D4 gives the condition on equal shares in Shannon's theorem, which then fixes F up to a positive factor. D2, applied to profiles with two targets above τ^*, then requires the ratio of their weights to stay the same when the gate moves without passing a value. The ratio can therefore depend only on the difference between their readouts. Only exponential and constant h have this property. For example, the entropy uses weights $\max(x_i - \beta, 0)$, with β as the gate. Consider two targets 1.0 and 0.5 units above β. Lowering β by 0.5 changes their weight ratio from 2 to 1.5.

This result requires weights to depend only on distance above the gate. Weights measured from a separate fixed reference satisfy D2 for any h. Rényi entropies of order other than 1 lack the grouping property, so they are excluded. Among scores of this form, the gated entropies satisfy D3 exactly when $T \geq \tau^* - x_0$ (Propositions 1 and 2). The equal-weight limit $T \to \infty$ also satisfies D3.

To compare how evenly weight is spread on the same panel, we sum the j largest entries of each set of shares. We call shares $\mathbf{q}$ *more even* than shares $\mathbf{p}$ if this sum is no larger for $\mathbf{q}$ than for $\mathbf{p}$, for every j.

Proposition 3 (broadening). On a fixed panel, let a profile have positive weights $w_1, \ldots, w_r$ ($r \geq 1$) and zero weights elsewhere. Let a score be a function f of the profile's weight shares. Require $f(\mathbf{q}) \leq f(\mathbf{p})$ whenever $\mathbf{q}$ is more even than $\mathbf{p}$. Changing one zero weight to a with $0 < a \leq \min_i w_i$ does not raise f.

The proof (Supporting Information, Section S4) shows that the added weight, being the smallest, makes the new shares more even than the old ones. The negative Shannon and

Rényi entropies, the largest share and the Gini coefficient all have this property. For the entropy, gated entropy, $P_{\max}$ and Gini, weights increase with affinity. Proposition 3 therefore means that a floor-coded pair can gain a positive weight without raising the score, provided it binds no more strongly than the weakest weighted target. The condition on a is needed: adding a target tied with the strongest can lower the entropy.

Recovery from Smaller Panels

We measure how well random smaller panels recover each score's full-panel ranking and how many compounds remain scorable.

We drew 50 subpanels of m targets uniformly at random from the curated panel at each tested size. We tested m = 10, 20, 30, 40, 50, 65, 80, 100, 125, 150, 200, 250 and 300, using only sizes below the full panel. Every score in Table 2, including the comparators, was computed on the same subpanels, with $\tau = \tau^*$ for the S-score, $\beta = x_0$ for the entropy and Gini, and $k = 1$ and $b = x_0$ for the ratio. We applied D1 to every score on every subpanel.

Coverage is the share of compounds scored on the full panel that remain scorable on a subpanel. We average it over the 50 subpanels of each size. Within a subpanel, it is the same for every score because D1 makes a compound eligible for all scores or none.

For *recoverability*, we first select clearly separated pairs from each score's full-panel ranking. Their rank gap must be at least a quarter of the number of scored compounds. Recoverability is the fraction of these pairs that remain scorable on a subpanel and keep their order. A pair tied on the subpanel counts as one half.

Conditional agreement is the Spearman correlation between subpanel and full-panel scores among compounds scored on both. We write p^* for the smallest tested panel size at which mean conditional agreement exceeds 0.90. This is a descriptive benchmark that we chose.

We also test whether compounds whose score depends heavily on one target are harder to recover. For each compound, we compute its mean absolute rank change at 50 targets, using only subpanels on which it remains scorable. Ranks are computed among compounds

scored on both panels and divided by their number. We also measure the largest change in the full-panel score when one target is left out. We report the Spearman correlation between these two quantities among scored compounds.

Supporting Information (Section S5) also reports recovery without D1, with every compound scored. These results depend partly on the conventions for undefined scores.

Results

Analysis Choices Change Agreement Between Scores

Among compounds with an active kinase, the S-score, entropy and Gini agree more with one another than with the ratio when the entropy and Gini use log-affinity weights (Table 3). We rank these compounds among themselves. On Davis, every compound has an active kinase. The three scores correlate at 0.963–0.999, while their correlations with the ratio are 0.51–0.56. Among the 200 active Klaeger compounds, the corresponding ranges are 0.89–0.997 and 0.38–0.64.

Table 3: Spearman correlations between median rankings. S is the S-score, E the entropy, G the Gini and R the ratio, with the entropy and Gini in their log-affinity forms. E_{K_a}: the published entropy, with floor-coded pairs given zero weight and, in parentheses, kept at the floor. In the active columns, compounds are ranked among themselves. Every Davis compound has an active kinase.

	Davis	Klaeger	
Pair	all ($n = 68$)	all ($n = 222$)	active ($n = 200$)
S–E	0.963	0.896	0.897
S–G	0.963	0.508	0.892
E–G	0.999	0.658	0.997
S–R	0.563	0.078	0.378
E–R	0.513	0.344	0.622
G–R	0.510	0.655	0.635
E_{K_a}–S	0.738 (0.344)	0.571 (−0.597)	0.564 (−0.412)
E_{K_a}–G	0.736 (0.368)	0.530 (0.077)	0.782 (−0.094)
E_{K_a}–R	0.861 (0.688)	0.657 (0.648)	0.916 (0.591)

Weighting. The published entropy uses weights inversely proportional to binding constants. It correlates more with the ratio than with the S-score or Gini on Davis, Klaeger and Metz. This holds under both treatments of floor-coded pairs (Table 3 and Supporting Information, Table S2). On Anastassiadis, the published percent-inhibition Gini correlates with the entropy at 0.63, and with the S-score and the ratio at only 0.34 and 0.33.

The published entropy's agreement with the S-score and Gini also depends on how floor-coded pairs are treated. These pairs make up 71% of the Davis matrix. Keeping them at the floor gives a correlation of 0.34 between the published entropy and S-score; giving them zero weight raises it to 0.74. The published and log-affinity entropies correlate at 0.37 on Davis and -0.31 on Klaeger when floor-coded pairs stay at the floor. With zero weight, these correlations rise to 0.74 and 0.79 (Figure 1). The negative Klaeger correlation comes from the floor-coded pairs. Keeping these pairs at the floor also causes the published entropy to fail D4 ("Criteria and the Gated Entropy").

Using all measured Klaeger affinities at their measured values, including the 194 weaker than the floor, barely changes the published entropy's ranking. Among active compounds, its Spearman correlation with the ranking that gives floor-coded pairs zero weight is 0.9996. Its correlations and recovery change by at most 0.02 (Supporting Information, Section S2).

Compounds ranked. Twenty-two Klaeger compounds have no target above the activity threshold. Including them makes the Gini correlate more with the ratio (0.66) than with the S-score (0.51). The difference is not significant: its bootstrap 95% interval is $[-0.35, 0.06]$, compared with $[0.16, 0.36]$ among active compounds. Each definition scores these compounds by its own convention. The S-score and entropy rank the 22 compounds among the most selective (median ranks 24 and 20 of 222). The ratio ranks them near the bottom (182). The Gini ranks them near the top when its score is defined and near the bottom when it is not (mean ranks 32 and 212). Their aggregated rank instability is $\sigma = 77.9$, compared with 32.2 for active compounds ($p < 0.001$, Mann–Whitney U test).

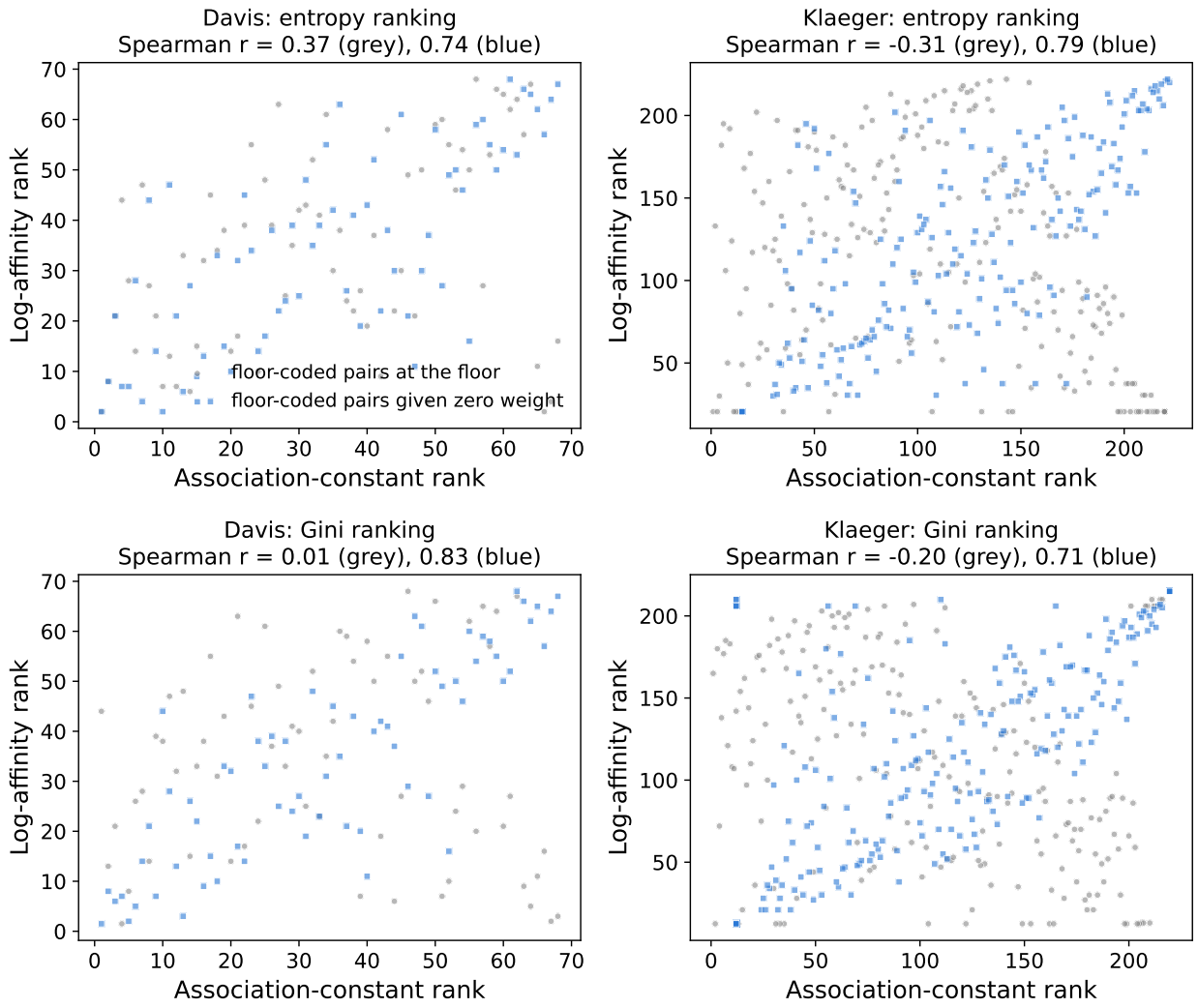


Figure 1: Rankings using weights inversely proportional to binding constants (x-axis) against the baseline-subtracted log-affinity form (y-axis, median rank over the baseline sweep), on Davis and Klaeger. Top: the published association-constant entropy. Bottom: the association-constant Gini (K_a-Gini) of Uitdehaag and Zaman (2011), distinct from Graczyk's percent-inhibition Gini. Grey: floor-coded pairs kept at the floor. Blue: floor-coded pairs given zero weight. Spearman r above each panel.

Within a single definition's sweep, their rank instability is significantly higher than that of active compounds only for the Gini (65.0 against 16.3). The difference reverses for the S-score (19.0 against 25.7, $p = 0.01$), whose score is 0 for all of these compounds at thresholds of 6 or above.

Sixteen of the 22 bind some kinase above the floor. Using only configurations with defined scores, these 16 have lower mean instability than active compounds under both the entropy and Gini. For the entropy, the means are 7.0 against 13.5; for the Gini, they are 6.7 against 13.6. Both comparisons give $p = 0.002$. The other six show no binding to any curated target, although all are established inhibitors. For three of them, the target was quantified for other compounds (Supporting Information, Section S3). Whether a compound is eligible for a selectivity score therefore depends on the panel and the threshold.

Agreement within each dataset. Among compounds with an active kinase, the S-score, entropy and Gini correlate at 0.81–0.999 across the four datasets. Each correlates less with the ratio (Supporting Information, Figure S1). On Anastassiadis, the separation is weak: the ratio correlates with the other three at 0.79–0.84, against 0.87–0.999 among those three. Including all compounds lowers the ratio's correlations with the other three scores to 0.48–0.55, because 45 compounds have no active kinase.

On Metz, the ratio correlates with the other three at only 0.15–0.20, because the other three largely follow how many kinases each compound was tested against. Untested pairs are coded at the floor, and nearly every tested value (99.9%) lies above it. A compound tested against more kinases therefore has more kinases with positive weight and looks less selective under the S-score, entropy and Gini. The ratio uses only the strongest value and one of the next five, so it is barely affected. The Spearman correlation between a compound's coverage and its median rank (rank 1 = most selective) is 0.98 for the entropy, 0.95 for the Gini, 0.76 for the S-score and 0.11 for the ratio. Compounds measured against more kinases are also active against a larger share of those measured, but that association is much weaker

($r = 0.33$).

Using mean rather than median ranks changes correlations among all compounds by at most 0.08 (Supporting Information, Section S2).

Target selection and filtering. On the released Davis panel, the ratio agrees less with the other definitions than on the curated panel (0.34–0.46 against 0.51–0.56, Supporting Information, Section S1). Restoring the low-confidence Klaeger pairs shifts the all-compound Klaeger correlations by at most 0.08 and correlations among the 200 compounds active under both filters by about 0.006. Restoring the fits makes two additional compounds eligible, increasing the active population to 202. Every p^* in the recovery analysis remains unchanged (Supporting Information, Section S1).

Changing the dataset vs. changing the score. We compared the 46 compounds profiled in both Davis and Klaeger on the 201 kinases shared by the panels. For the same definition across datasets, rankings correlate at 0.34–0.84 (mean 0.61). For different definitions within one dataset, they correlate at -0.08–1.00 (mean 0.57). Among the 40 compounds active in both datasets, the means are 0.72 and 0.73, and the Gini's cross-dataset correlation rises from 0.45 to 0.84. On these compounds, the choice of definition moves rankings about as much as the change of dataset. On average, the 46 compounds are active against 15.0% of the Davis panel and 3.9% of the Klaeger panel. Without restricting the panels to their shared kinases, the cross-dataset correlations are lower (mean 0.53).

Parameter Sensitivity and Profile Associations

Moving each definition's activity boundary across B reorders a substantial share of compound pairs. The S-score's threshold reverses 8.4–13.0% of pairs of scored compounds on Davis, Klaeger and Anastassiadis, and 21.8% on Metz. The entropy's and Gini's baselines reverse 7.9–12.4% on the same three datasets, and 22.5–22.8% on Metz. For the ratio, b is a lower bound on the reference off-target affinity used to calculate the score. Varying this bound

reverses 3.0–13.7% of pairs on Klaeger and 2.1–9.7% on Anastassiadis, with the fraction rising as k increases. The ranges are 0–2.6% on Davis and 0.04–1.03% on Metz.

Sweeping the gated entropy's gate, with T fixed at $\tau^* - x_0$, reverses 9.2–14.6% of pairs on Davis, Klaeger and Anastassiadis, and 23.8% on Metz. On Klaeger, the per-compound instability under the gate sweep is close to that under the entropy's baseline sweep (mean rank $\sigma = 6.7$ and 7.0, Supporting Information, Figure S2). Supporting Information (Table S3) gives every value.

Reference off-target. When an active compound's two best-bound kinases have near-identical affinities, the ratio depends heavily on k. $R(1)$ compares the first kinase with the second, treating the second as an off-target (a kinase the compound is not meant to bind). For near-tied kinases this difference is close to zero, so the compound scores as non-selective. $R(2)$ compares the first kinase with the third, treating the second as a co-target (a second intended target). The same compound can then score as highly selective. Among the 22 active Klaeger compounds with a top_1–top_2 gap below 0.05, the means are $R(1) = 0.02$ and $R(2) = 0.50$. The appropriate reference depends on whether the second kinase is an intended target. The profile alone does not establish this. On Davis, fedratinib, a JAK2 inhibitor, has the most unstable ratio ranking. JAK2 and GAK are tied among its 123 active targets (Figure 2, left).

Profile associations. Among active compounds, three associations between profile features and rank instability are significant at $p < 0.05$ on both pK_d datasets before correction for multiple tests (Supporting Information, Table S4). The number of active kinases correlates negatively with entropy instability on every dataset ($r = -0.30$ to -0.42). On Davis, this association is only nominally significant ($p = 0.01$). The number of active kinases also correlates with Gini instability on Davis and Klaeger ($r = -0.31$ and -0.25). On Klaeger, entropy instability correlates with the share of a compound's above-floor kinases that lie within the baseline sweep ($r = 0.22$, $p = 0.002$). The top_1–top_2 gap correlates with ratio

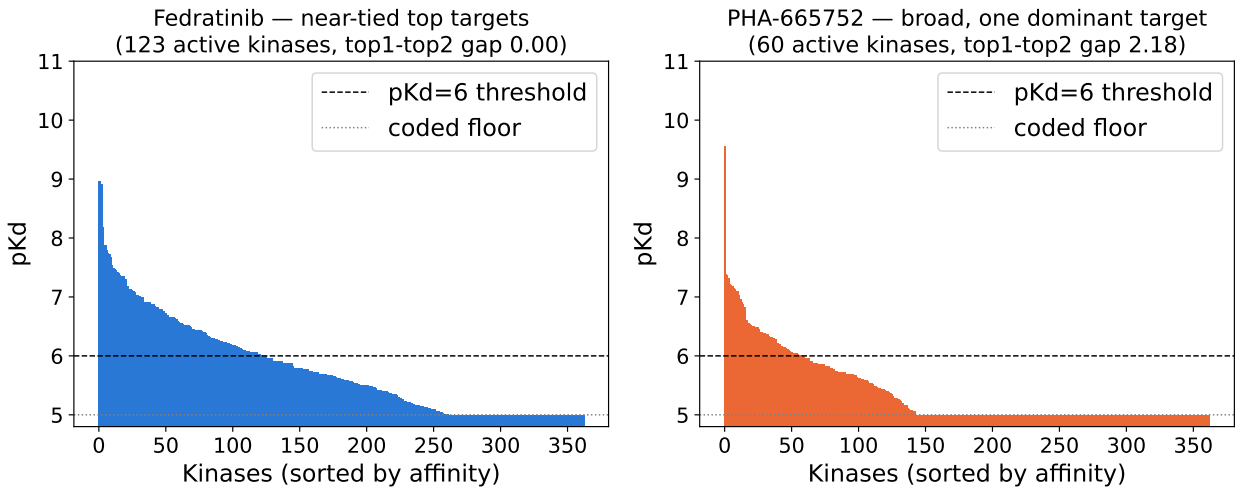


Figure 2: Binding profiles of two Davis compounds, targets sorted by affinity. Left: fedratinib. Right: PHA-665752. Dashed line: p$K_d = 6$ activity threshold. Dotted line: p$K_d = 5$ coded floor.

instability on Davis, Klaeger and Metz ($r = -0.44$, -0.28 and -0.26, all $p < 0.001$). It does not correlate on Anastassiadis ($r = 0.00$). Excluding eight compounds whose two best targets are at the 100% inhibition ceiling gives -0.15 with $p = 0.10$. After Bonferroni correction, only the ratio–gap association remains significant on both pK_d datasets. The Holm and Benjamini–Hochberg procedures give the same result.

On Klaeger, but not Davis, S-score instability also correlates with the range of active affinities ($r = -0.32$). The partial correlation is $r = -0.22$ with the number of active kinases held fixed. When all 222 Klaeger compounds are ranked together, the skewness of the active affinities is the feature most strongly associated with Gini instability ($r = +0.26$). Among active compounds, this association is not significant ($r = +0.07$, $p = 0.31$). The association in the full set therefore comes mainly from how compounds with no active kinase are scored. A weaker association remains when the number of active kinases is held fixed (Supporting Information, Section S3).

Broad profiles. Among active compounds, seven of the ten largest entropy–ratio disagreements on Klaeger involve profiles with 14–27 active kinases. Six of these have one clearly dominant target (top_1–top_2 gap above 1). The ratio ranks these seven 31st to 63rd of 200,

while the entropy and Gini rank them 146th to 180th. The ratio reflects the dominant target; the entropy and Gini reflect the number of active kinases. Bosc et al.[8] attributed disagreements for individual compounds to such broad profiles with one dominant target. On Davis, PHA-665752 has the highest aggregated rank instability ($\sigma = 16.4$). It has one dominant target among 60 active targets (Figure 2, right).

Criteria and the Gated Entropy

Table 4 shows whether each score meets D1 to D4, judged from its definition (Methods). None of the four published definitions withholds scores from compounds with no active kinase, so none meets D1. Any score can adopt this eligibility rule; the comparators and gated entropy do so here.

The baseline-subtracted entropy and Gini violate D2, as does the ratio. The S-score, active count and gated entropy satisfy it. The published entropy and $P_{\max}$ have no boundary parameter, so D2 does not apply to them. The S-score with $\tau > x_0$, the entropy with $\beta = x_0$, the Gini and the published entropy violate D3. The other scores, including $P_{\max}$, satisfy it. Every score in the table satisfies D4, as do $P_{\max}$ and the published entropy with floor-coded pairs given zero weight.

D2 does not make a ranking independent of the boundary. For the ratio, the largest fraction of reversed pairs across datasets is 3.0% at $k = 1$, rising to 13.7% at $k = 5$ (Supporting Information, Table S3).

Table 4: Evaluation of the score implementations against D1–D4. Count: negated number of active targets. Entropy and Gini use baseline-subtracted weights. ✓: criterion satisfied. ×: criterion violated.

	S-score		Count	Entropy		Gini	Ratio	Gated
	$\tau > x_0$	$\tau = x_0$		$\beta = x_0$	$\beta = \tau^*$	$\beta \geq x_0$		entropy
D1 (eligibility)	×	✓	✓	×	✓	×	×	✓
D2 (locality)	✓	✓	✓	×	×	×	×	✓
D3 (monotonicity)	×	✓	✓	×	✓	×	✓	✓
D4 (orientation)	✓	✓	✓	✓	✓	✓	✓	✓

From the score definitions. The S-score and active count depend on the boundary only through which targets lie above it, so they satisfy D2. The entropy and Gini subtract β from every readout above it. The ratio $R_b(k)$ uses b as a lower bound on the reference off-target value. These scores can therefore change even when the boundary passes no target value, violating D2. Moving the gated entropy's gate without crossing a value multiplies all weights above it by the same factor. The factor cancels in the weight shares, so D2 holds.

The S-score violates D3 for any threshold above the floor. Adding a kinase bound between the floor and threshold increases N without adding a hit, making the compound look more selective. With the threshold at the floor, every addition allowed by D3 is a hit. A compound with C hits among N kinases then moves from $-C/N$ to $-(C+1)/(N+1)$, which is not higher. The active count gives the same ranking as the S-score at τ^* on any one panel, but does not divide by N. A kinase added at or below τ^* does not change the count, so D3 holds.

The Gini violates D3 because a weakly bound kinase can increase inequality. The ratio satisfies D3 because a weak addition cannot displace the primary target or lower the reference off-target. For $P_{\max}$, a weak addition leaves the numerator unchanged and raises the denominator, so D3 also holds. The entropy satisfies D3 at $\beta = \tau^*$, where weak binding has zero weight. The gated entropy satisfies D1 through its eligibility rule and D3 by Proposition 1 in Methods.

D3 concerns adding a target. Strengthening binding to an existing target can raise the gated entropy's score. For example, with one kinase at $\mathrm{p}K_d = 7.0$ and fifty at 5.1, the score is -5.502. Raising one of the fifty to 6.0 raises it to -5.491.

D4 requires that binding one more kinase as strongly as the others must not raise the score. Consider a compound that binds m kinases at the same value a, with the rest of the panel at the floor. Methods denotes these profiles by $\mathbf{v}_m$. The scores are:

- active count: $-m$
- S-score with $\tau < a$: $-m/N$

- the entropies and the gated entropy, whose m weights are equal: $-\log_2 m$
- $P_{\max}$: $1/m$
- Gini: $(N-m)/N$
- ratio $R_b(k)$: $a - \max(x_0, b)$ for $m \leq k$ and 0 otherwise.

None of these values rises as m grows, so all of these scores satisfy D4. These values assume zero weight for floor-coded pairs. For the entropy and Gini, this requires $\beta \geq x_0$. On Anastassiadis, their sweeps also include baselines below the 20% floor, where both fail D4. For example, on its 291 kinases, with $\beta = 0\%$ and targets at 60%, a second target raises the Gini from 0.0068 to 0.0135.

Keeping floor-coded pairs at the floor also causes the published entropy to fail D4 at the panel sizes used here. On 363 kinases with targets at $\mathrm{p}K_d = 7$, a second target raises its score from -7.41 to -6.76 bits. The largest share falls as m grows, so $P_{\max}$ satisfies D4 under either floor treatment.

D1 to D3 alone do not fix the direction of a score. Reversing the active count's sign makes a compound look more selective as it binds more active kinases. This score depends only on which targets lie above τ^* and does not change when a kinase is added at or below τ^*. It therefore satisfies D1 to D3. Yet it rises with the number of equally strong targets ($+m$ on $\mathbf{v}_m$), violating D4. The gated entropy without its non-binder gate also fails D4, since floor-coded pairs carry weight, so adding equally strong targets can concentrate the distribution.

For the entropy, gated entropy, $P_{\max}$ and Gini, Proposition 3 gives a stronger result than D4. Turning a floor-coded pair into a target no stronger than the weakest weighted target cannot raise the score. This holds for any profile, including those with unequal binding strengths.

Counterexamples from the definitions. The published entropy violates D3 even when floor-coded pairs have zero weight. A profile with one target at $\mathrm{p}K_d = 6.01$ and fifty at 5.01

scores -5.35 bits. Adding one at 6.0 raises the score to -5.19.

The baseline-subtracted entropy violates D3 at $\beta < \tau^*$. Its hard cutoff gives a kinase just above β a weight close to zero. A profile with many such kinases therefore has high entropy. Adding one kinase bound more strongly, but still below τ^*, concentrates the distribution. With $\beta = 5.0$, a profile with one kinase at $\mathrm{p}K_d = 7.0$ and fifty at 5.1 scores -4.89 bits. Adding one at 6.0 raises the score to -4.83. Adding the kinase at 6.0 lowers the gated entropy's score from -5.502 to -5.522. The exponential weights give that kinase only 2.5 times the weight of a kinase at 5.1 (2.72 against 1.11). The entropy's hard cutoff gives it 10 times the weight (1.0 against 0.1).

Empirical checks. We also tested D3 for the entropy and gated entropy on measured profiles of active compounds. Adding a hypothetical kinase at τ^* never raises the entropy's score on Davis, Klaeger or Metz at any baseline in their sweeps. On Anastassiadis, it can raise the score at baselines below the 20% floor. This occurs for 33% of compounds at $\beta = 0\%$ and 16% at 5%. It never raises the gated entropy's score on any dataset.

We also added measured kinases bound below the threshold to random 50- or 100-kinase subpanels of the same profiles. This raises the entropy's score in at most 0.1% of cases on any dataset and never raises the gated entropy's score.

Across the four datasets, the gated entropy's score falls as the number of active kinases rises (Spearman -0.76 to -0.95). Without the non-binder gate, the correlation reverses ($r = +0.67$ on Klaeger). Its rankings agree with the entropy's (0.969–0.993) and with those of the S-score with $\tau = x_0$ (0.964–0.996). This S-score also satisfies D1 to D4.

Moving τ^* from 5.5 to 7 (40% to 70% on Anastassiadis), with $T = \tau^* - x_0$ changed with it, affects both eligibility and rankings. Among compounds scored at both values, the Spearman correlation with the default ranking is 0.85 to 0.999. The lowest values are at $\tau^* = 5.5$ on Davis and Klaeger, where T falls to 0.5 pK units.

Adding random noise to the measured values changed which compounds passed D1, as described in Methods. At most 11 compounds changed status in any draw. Among

compounds scored both with and without noise, the gated entropy's ranking correlated with its noise-free ranking at Spearman 0.972 or higher. For the entropy, the correlation was 0.983 or higher.

On these datasets, the gated entropy does not improve ranking recovery over simpler scores.

What the criteria determine. The active count, the S-score with $\tau = x_0$ and the gated entropy satisfy all four criteria. Within the class defined in Proposition 2, D2 and D4 allow only gated entropies or equal weights. The equal-weight score is $-\log_2 n$ for n targets above the gate (the limit $T \to \infty$). D3 further restricts the gated entropies to $T \geq \tau^* - x_0$ (Proposition 1); the equal-weight score also satisfies it. On Klaeger, rankings for $T = 1$, 1.5 and 2 agree at Spearman 0.985 or higher. Supporting Information, Section S4, compares other variants.

Recovery from Small Panels

On small random panels, some compounds can no longer be scored. At 50 targets, 88% of the compounds scored on the full panel are still scored on Davis, 71% on Klaeger, 77% on Anastassiadis and 99% on Metz (Figure 3, top). Compounds with few active targets are most often lost. On a 50-target Klaeger subpanel, a compound with one active target remains scorable with probability 0.23. With ten active targets, the probability is 0.93. Rank agreement among compounds that remain scorable can therefore be high even when many compounds with few active targets are lost.

Coverage limits how many pairs can retain their order. To keep 90% of clearly separated pairs, most scores need 150 targets on Davis, 200 on Klaeger and Anastassiadis, and 40 on Metz. The ratio, published entropy and $P_{\max}$ need as many targets or more (Figure 3, middle).

Among compounds scored on both panels, scores whose rankings depend mostly on the few strongest targets recover slowest (Figure 3, bottom). The S-score with $\tau = x_0$, the

entropy with $\beta = x_0$ and the Gini rank compounds almost as the share of panel entries above the operational floor does (Spearman above 0.97 with the S-score at $\tau = x_0$ on every dataset). We call these three the *share-following scores*. The entropy and Gini curves differ from the S-score's by at most 0.012. Figure 3 therefore shows only the S-score with $\tau = x_0$ for this group.

The share-following scores recover fastest on Davis, Klaeger and Anastassiadis, reaching p^* at 20, 50 and 30 targets. The S-score at τ^* and active count need 30, 80 and 50 targets. These two give identical rankings on any one panel and share a curve in Figure 3. The entropy with $\beta = \tau^*$ needs 40, 100 and 65 targets.

On Metz, where untested pairs are set to the floor, the S-score at τ^*, active count and entropy with $\beta = \tau^*$ recover fastest. They need 50 targets, compared with 65 for the share-following scores. Using benchmarks of 0.80, 0.85 or 0.95 instead of 0.90 preserves these comparisons: the fastest scores reach the benchmark no later than any score outside their group.

The published entropy and $P_{\max}$, with floor-coded pairs given zero weight, need 300 of 363 targets on Davis and 200 of 217 on Klaeger. On Metz, the published entropy needs 150 of 172 targets; $P_{\max}$ does not reach the benchmark at the largest subpanel tested (150). The ratio needs 200 of 217 targets on Klaeger. It does not reach the benchmark at the largest subpanel tested on Davis (300 of 363), Anastassiadis (250 of 291) or Metz (150 of 172). The published entropy gives weak binding a small positive weight. Including weak binding therefore does not by itself ensure fast recovery.

The gated entropy follows the share of entries above the operational floor less closely (Spearman 0.964–0.996 with the S-score at $\tau = x_0$). Its mean Spearman curve lies below that of the S-score with $\tau = x_0$ on every dataset, by up to 0.069. It reaches p^* at 50, 65, 30 and 80 targets on Davis, Klaeger, Anastassiadis and Metz.

The size p^* is defined by mean agreement across subpanels. At p^*, 62–90% of individual subpanels exceed 0.90 for the share-following scores (Supporting Information, Section S5).

Compounds whose score changes most when one target is left out tend to have larger rank errors at 50 targets on Davis, Anastassiadis and Metz. For the entropy with $\beta = x_0$, Gini and gated entropy, the Spearman correlations are $r = 0.18$–0.70. On Klaeger, this association is weak ($r \leq 0.13$).

Without D1, the Gini recovers as fast as the entropy when compounds with undefined Gini values are excluded (Supporting Information, Section S5). These compounds are otherwise ranked as least selective.

Scale of the Gated Entropy and Recovery

The gated entropy satisfies D1 to D4 for any scale $T \geq \tau^* - x_0$. At $T = 2(\tau^* - x_0)$, it reaches p^* at 20, 50, 30 and 65 targets on Davis, Klaeger, Anastassiadis and Metz. These are the same sizes needed by the share-following scores, compared with 50, 65, 30 and 80 at $T = \tau^* - x_0$. A larger scale reduces weight differences between targets. The rankings then approach those of the S-score with $\tau = x_0$ (Spearman 0.9968–0.9996).

Increasing the scale changes the full-panel ranking as well as its recovery from smaller panels. Faster recovery shows that the new ranking is easier to reproduce, but it does not establish that it better represents selectivity. The other analyses use $T = \tau^* - x_0$, the smallest allowed scale and the one most sensitive to affinity differences. Supporting Information, Section S5, gives the same comparison for $T = 4(\tau^* - x_0)$.

Discussion

Discrepancies between the rankings of the scores compared here depend on protocol choices made before any score is computed. Moreover, the different ranking computations summarize different aspects of a binding profile. The S-score, entropy and Gini agree more with one another than with the ratio when the entropy and Gini are computed from readouts above a common baseline and compounds without an active kinase are left out. With weights

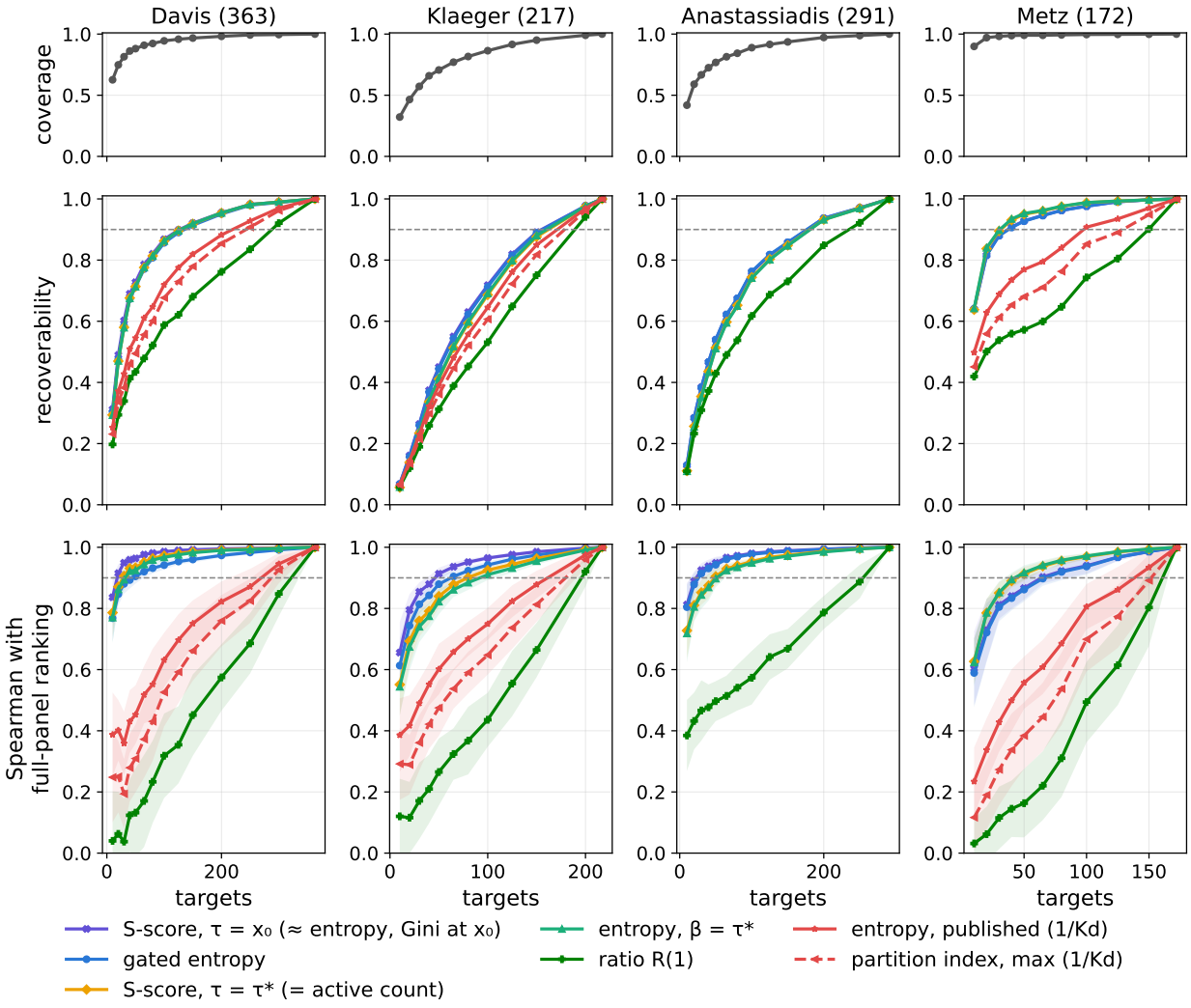


Figure 3: Recovery from random subpanels, with D1 applied to all scores (50 subpanels per size). Top: share of compounds scored on the full panel that remain scorable. Middle: share of clearly separated full-panel pairs retained in order. Bottom: mean Spearman correlation with each score's full-panel ranking among compounds scored on both panels. Shading: ±1 standard deviation. Dashed line: 0.90.

inversely proportional to the binding constants, or with compounds that have no active kinase included, this grouping weakens or disappears. When most entries are coded at the floor, as on Metz, the S-score, entropy and Gini agree for a different reason: each ranks compounds largely by how many kinases the compound was tested against. Bosc et al.'s window score correlates strongly with the ratio (0.91–0.96), and their ranking score is a ratio by definition. A proxy for the CATDS score of Klaeger et al., the top target's share of the profile, instead correlates more with the Gini than with the ratio on every dataset (Supporting Information, Section S6). The ratio and the S-score, entropy and Gini disagree most on broad profiles with one dominant target. On compounds and kinases shared by two datasets, changing the definition moved rankings about as much as changing the dataset.

What the Design Criteria Guarantee

D1 is a reporting rule that any score can adopt. D2 and D3 test how a score changes when the boundary moves and when a weakly bound target is added. The baseline-subtracted entropy and Gini fail D2, as does the ratio used here. The published entropy has no boundary parameter, so D2 does not apply to it. The S-score fails D3 when its threshold is above the floor. The entropy with $\beta = x_0$, the Gini, and the published entropy also fail D3. D4 fixes the direction of a score, which D1 to D3 leave open.

Meeting D2 does not make rankings independent of the boundary, so the boundary must still be fixed by protocol. D3 concerns adding a weakly bound kinase, above the floor and at or below the activity threshold. A kinase added at the floor has zero weight in the gated entropy and leaves its score unchanged. Adding a kinase at or below the activity threshold cannot raise the score; most entries fall in this range. Strengthening binding to an existing target can raise it.

D4 applies only to profiles with equally strong targets. With unequal binding strengths, a compound can bind more targets yet have most of its binding concentrated on a few. A score that measures concentration can then rank it as more selective without violating

D4. Correlation with the number of active kinases on real data therefore does not establish whether a score satisfies D4.

Among the scores studied, only the gated entropy meets all four and weights targets by affinity. Within the class defined in Proposition 2, the criteria allow exponential weights and their equal-weight limit. For finite scales, D3 requires the scale to be at least the distance between the floor and the activity threshold. On the affinity datasets, the published entropy with floor-coded pairs given zero weight has a smaller scale and violates D3.

Two simpler scores give almost the same rankings as the gated entropy: the entropy with $\beta = x_0$ and the S-score with $\tau = x_0$. The entropy weights targets by affinity but fails D2 and D3. The S-score meets all four criteria but weights every target above the operational floor equally. Increasing the gated entropy's scale reduces weight differences and brings its rankings closer to those of this S-score. At twice the smallest allowed scale, the gated entropy retains its affinity weights and recovers rankings from small panels as fast as the S-score, but does not surpass it.

Reporting Recommendations

Each recommendation follows from a result above.

1. State which targets the score is computed over, including how the panel was curated, and the score implementation, including the weighting and its parameters, such as a baseline or the gated entropy's scale.

2. Distinguish measured affinities, pairs with no detected binding, rejected fits and untested pairs, and state the value and weight each receives.

3. Apply D1: state the activity threshold, withhold or flag the scores of compounds with no target above it, and rank the eligible compounds among themselves.

4. Fix the activity boundary in the analysis protocol and report how rankings change

when it is moved, for example by each compound's rank instability across the sweep. Assess agreement across datasets individually.

5. For small panels, report how many compounds remain scorable as well as how well the ranking of those compounds is recovered.

6. When comparing studies that use different scores, fix the comparison protocol in advance and report the compounds on which the scores disagree.

Limitations

The four datasets, from three assay technologies, cover 68 to 704 compounds and 172 to 363 kinases, and the results may not hold for other compounds or panels. The datasets also differ in how many interactions they find: the 46 compounds profiled on both Davis and Klaeger are active against 15.0% of the Davis panel and 3.9% of the Klaeger panel on average. On Davis, Klaeger and Anastassiadis every compound was tested against the whole panel, and only 1.1% of the Anastassiadis entries are missing (set to 0% inhibition). On Metz, each compound was tested against some particular subset of the 172 kinases.

On no dataset is the floor a measured detection limit. It codes non-detection (on Davis), non-detection, low-confidence fits and weak measured affinities (on Klaeger), and untested pairs (on Metz). On Anastassiadis, it is an operational cutoff for weak inhibition. Fixing the boundary at the floor therefore depends on how each dataset codes these pairs. Random subpanels are one sampling design, so panels chosen by kinase family, as many screening panels are, may recover rankings differently. Cell-free panels can also miss interactions that cellular target-engagement assays detect[19,20].

No score considered here uses target identity or exposure. Two compounds with the same affinity distribution receive the same score whether their off-targets are inert or linked to a class toxicity, and an affinity alone does not establish whether a target is occupied at a tolerated dose. A selectivity score is at best a summary of the binding profile, to be weighed

with target-specific liability knowledge and exposure modeling. In the exploratory analysis of 46 approved Klaeger compounds, we detected no statistically significant correlations between selectivity rankings and the proportions of their adverse-event reports that were serious, fatal or led to hospitalization (Supporting Information, Section S7). This result does not establish an absence of clinically relevant associations.

Conclusion

Kinase inhibitor selectivity rankings depend on choices usually left implicit: how affinities are weighted, which compounds are ranked, how floor-coded pairs are treated, which targets the panel includes, and where the activity boundary is placed. These choices change whether the common scores agree, how much a ranking moves when the activity boundary moves, and how much of a ranking a small panel retains. Selectivity analyses should state them, apply an activity-eligibility rule, fix the activity boundary in advance and report its effect, as well as report how many compounds remain scorable on small panels. Rank agreement can remain high even when compounds with few active targets become unscorable on a smaller panel.

The gated entropy meets all four design criteria while weighting targets by binding strength, including weak binding above the floor. Under the assumptions of Proposition 2, the criteria allow only exponential or equal weights. For exponential weights, criterion D3 sets the minimum scale at which adding a weakly bound target cannot make a compound appear more selective. On the datasets analyzed here (Davis, Klaeger, Anastassiadis and Metz), it has no recovery advantage over simpler scores. Any scale at or above this minimum satisfies the criteria. Increasing the scale reduces weight differences between targets and changes both rankings and recovery from small panels, so the scale must be stated when the score is reported.

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Supporting Information

Sections S1–S8 of the Supporting Information, with Tables S1–S9 and Figures S1–S4, give the full results for each analysis, the proofs of Propositions 1–3 and the script behind every reported number.

Data and Software Availability

All data and software used in this study are publicly available. The Davis kinase inhibitor profiling dataset is available at `https://github.com/dingyan20/Davis-Dataset-for-DTA-Prediction`. The Klaeger et al. dataset is available as supplementary data to the original publication (Science, 2017, `https://doi.org/10.1126/science.aan4368`). The Anastassiadis et al. (*Nat. Biotechnol.* 2011) and Metz et al. (*Nat. Chem. Biol.*

2011) datasets are available as supplementary data to their original publications (`https://doi.org/10.1038/nbt.2017`; `https://doi.org/10.1038/nchembio.530`). All analysis scripts, processed data files, and the LaTeX source for this manuscript are available at `https://github.com/polinavino/kinase-selectivity-definitions`; this repository includes the code and full per-definition results for the secondary clinical-outcome analysis (FDA Adverse Event Reporting System (FAERS) adverse-event report rates for the FDA-approved Klaeger compounds, obtained via openFDA queries). The clinical correlations use a frozen snapshot in `selectivity_outcomes_merged.csv`; Supporting Information (Section S7) documents its archival provenance and the query scope. The analyses used Python 3 with NumPy[21], pandas[22] and SciPy[23], and Supporting Information (Table S9) names the script behind every result. No confidential data were used in this study. All results reported in this manuscript are fully reproducible from the publicly available data and code. A preprint of this manuscript is available on ChemRxiv at `https://doi.org/10.26434/chemrxiv.15001618/v1`.

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Conflict of Interest

The author declares no conflict of interest.

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